



  **Jibola Ayomide Adeshola** ▾



Home > My courses > PHY-CM102_2024/2025 > General > CLICK HERE TO START PHY-CM 102/PHS 122 CA - GROUP D

Question 1

Not yet answered

Marked out of 1

When a glass rod is rubbed with silk, which of the following statements best describes what happens?

Select one:

- ☒ a. **electrons are removed from the rod**
- ☐ b. **the silk remains neutral**
- ☐ c. **protons are removed from the silk**
- ☐ d. **positive ions are removed from the silk**

Question 2

Not yet answered

Marked out of 1

A potentiometer is a device used to

Select one:

- ☒ a. **compare two voltages**
- ☐ b. **measure a current**
- ☐ c. **compare two currents**
- ☐ d. **measure a voltage**



Question 3

Not yet answered

Marked out of 1

The work done in moving a positive charge from one point to another in an equipotential plane is?

Select one:

- ☐ a. Maximum
- ☐ b. Positive
- ☐ c. Negative
- ☒ d. Zero

Question 4

Not yet answered

Marked out of 1

Direction of conventional current is

Select one:

- ☐ a. direction of neutron flow
- ☐ b. direction of electron flow
- ☒ c. direction of flow of positive charge
- ☐ d. same as that of electric current

Question 5

Not yet answered

Marked out of 1

A resistor, $100 \, \Omega$, is connected to an a.c. source given as $v = 300 \sin \omega t$. If $\omega = 1256 \text{ rad/s}$, find the power dissipated by the resistor.

Select one:

- ☐ a. 900W
- ☐ b. 700W
- ☒ c. 450W
- ☐ d. 300W



Question 6

Not yet answered

Marked out of 1

A capacitance carries a charge of 0.1c at 5v. What is its capacitance?

Select one:

- ☒ a. **0.02F**
- ☐ b. **0.2F**
- ☐ c. **0.5F**
- ☐ d. **5F**

Question 7

Not yet answered

Marked out of 1

Which of the following is correct about the acceleration of a positively charged particle?

Select one:

- ☐ a. The acceleration is against the direction of the electric field
- ☐ b. The acceleration is uniform
- ☒ c. The acceleration is in the direction of the electric field
- ☐ d. The acceleration is infinitely large

Question 8

Not yet answered

Marked out of 1

Whenever there is force on magnetic pole, there exists

Select one:

- ☒ a. **magnetic field**
- ☐ b. **electric field**
- ☐ c. **current**
- ☐ d. **voltage**



Question 9

Not yet answered

Marked out of 1

Which of the following is/are correct about a charge q , moving with velocity, v , in a uniform magnetic field?

I. The magnitude and direction of the force, $\mathbf{F}_B$ depends on $\mathbf{v}$ and $\mathbf{q}$;II. The magnitude and direction of the force, $\mathbf{F}_B$ independent of $\mathbf{v}$ and $\mathbf{q}$; III. The magnitude and direction of the force, $\mathbf{F}_B$ is zero always.

Select one:

- ☒ a. I only
- ☐ b. II only
- ☐ c. II and III
- ☐ d. I, II and III

Question 10

Not yet answered

Marked out of 1

Strength of magnetic field is known as

Select one:

- ☐ a. **flux**
- ☐ b. **density**
- ☐ c. **magnetic strength**
- ☒ d. **magnetic flux density**

Question 11

Not yet answered

Marked out of 1

A car battery is rated at 80 A · h. An ampere-hour is a unit of:

Select one:

- ☐ a. power
- ☐ b. energy
- ☐ c. current
- ☒ d. charge



Question 12

Not yet answered

Marked out of 1

A battery of fixed emf E and internal resistance r is connected to a variable external resistor R .
what would be the relation between R and r when maximum power is delivered.

Select one:

- ☐ a. $R=2r$
- ☒ b. $R=r$
- ☐ c. $R=r^2$
- ☐ d. $R=\sqrt{r}\sqrt{r}$

Question 13

Not yet answered

Marked out of 1

The sum of the voltages over any closed loop is equal to

Select one:

- ☒ a. 0V
- ☐ b. Infinity
- ☐ c. 1V
- ☐ d. 2V

Question 14

Not yet answered

Marked out of 1

In a series circuit, which of the parameters remain constant across all circuit elements such as resistor, capacitor and inductor etcetera?

Select one:

- ☐ a. Voltage
- ☒ b. Current
- ☐ c. Both voltage and current
- ☐ d. Neither voltage nor current



Question 15

Not yet answered

Marked out of 1

According to equation ' $r = (mv)/(Be)$ ', faster moving particles

Select one:

- ☐ a. move in smaller circle
- ☐ b. move straight
- ☒ c. move in bigger circle
- ☐ d. move randomly

◀ Announcements

Jump to...

